

Conversion Log

Production Conversion Log

June 2025 · Girardot, Colombia

Submission reference	RTF0-310825
Reporting period	June 2025 (2025-06)
Plant	Girardot, Colombia
Product	DEV-P100 — refined pyrolysis oil
Off-taker	Crown Oil Limited (UK)
Regulator	UK DfT — LCF Delivery Unit
Density — EU-grade	0,770 kg/L
Density — PLUS-grade	0,853 kg/L
Production days observed	23 of 30 (7 non-production days)
Source	daily_production table (DFT backend, snapshot 2026-05-21T00:00:00Z)
Generated	2026-05-21T00:00:00Z

Signed for and on behalf of OisteBio GmbH

Paolo Ughetti

Managing Director

Date: _____

Executive Summary

Per-day kilogram-to-litre conversion log for **June 2025** production at the Girardot, Colombia pyrolysis facility, operated by **OisteBio GmbH**. Output product is **DEV-P100 — refined pyrolysis oil**, supplied to **Crown Oil Limited (UK)** under the UK DfT — LCF Delivery Unit reporting framework. Litres are derived from kilograms via the nominal product densities listed below, as materialised by the `litres_eu` and `litres_plus` GENERATED ALWAYS columns on the `daily_production` table (Alembic migration `0007_persist_production_litres.py`).

Monthly aggregate

Metric	Value
Feedstock routed to thermal conversion (input)	3 063 732,000 kg
Total EU-grade production	851 717,496 kg / 1 106 126,621 L
Total PLUS-grade production	1 268 221,650 kg / 1 486 778,022 L
Liquid-product yield (EU + PLUS) over input	69,19 %
Grand total volume (EU + PLUS)	2 592 904,643 L

Co-products tracked separately on the same daily ledger (not converted to litres because they are solid / gaseous): carbon black **412 909,375 kg**, steel scrap **277 239,152 kg**, water condensate **50 138,995 kg**, syngas **140 094,252 kg**, recorded losses **63 411,080 kg**. Mass-balance closure across all streams is reported in Annex A.

Daily Conversion — June 2025

One row per production day from the `daily_production` table. Days with no production record (weekends, public holidays, maintenance) are listed in the gap line below. Litres computed at constant density: **EU 0,770 kg/L, PLUS 0,853 kg/L**. Liquid yield = (EU kg + PLUS kg) / input kg.

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-06-03	114 373,000	31 795,694	47 344,322	0,770	0,853	41 293,109	55 503,308	69,19 %	—
2025-06-04	125 009,000	34 752,502	51 747,059	0,770	0,853	45 133,119	60 664,782	69,19 %	—
2025-06-05	126 235,000	35 093,330	52 254,557	0,770	0,853	45 575,753	61 259,739	69,19 %	—
2025-06-06	135 562,000	37 686,236	56 115,438	0,770	0,853	48 943,164	65 785,977	69,19 %	—
Total	3 063 732,000	851 717,496	1 268 221,650	—	—	1 106 126,621	1 486 778,022	69,19 %	

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-06-07	116 822,000	32 476,516	48 358,077	0,770	0,853	42 177,294	56 691,767	69,19 %	—
2025-06-09	131 888,000	36 664,864	54 594,598	0,770	0,853	47 616,706	64 003,046	69,19 %	—
2025-06-10	129 893,000	36 110,254	53 768,774	0,770	0,853	46 896,434	63 034,905	69,19 %	—
2025-06-11	145 368,000	40 412,304	60 174,599	0,770	0,853	52 483,512	70 544,665	69,19 %	—
2025-06-12	134 101,000	37 280,078	55 510,662	0,770	0,853	48 415,686	65 076,978	69,19 %	—
2025-06-13	134 925,000	37 509,150	55 851,754	0,770	0,853	48 713,182	65 476,851	69,19 %	—
2025-06-14	124 780,000	34 688,840	51 652,265	0,770	0,853	45 050,442	60 553,652	69,19 %	—
2025-06-16	145 518,000	40 454,004	60 236,691	0,770	0,853	52 537,668	70 617,457	69,19 %	—
2025-06-17	131 381,000	36 523,918	54 384,728	0,770	0,853	47 433,660	63 757,008	69,19 %	—
2025-06-18	146 345,000	40 683,910	60 579,025	0,770	0,853	52 836,247	71 018,787	69,19 %	—
2025-06-19	141 654,000	39 379,812	58 637,201	0,770	0,853	51 142,613	68 742,322	69,19 %	—
2025-06-20	126 829,000	35 258,462	52 500,442	0,770	0,853	45 790,210	61 547,998	69,19 %	—
2025-06-21	134 046,000	37 264,788	55 487,895	0,770	0,853	48 395,829	65 050,287	69,19 %	—
2025-06-24	151 739,000	42 183,442	62 811,854	0,770	0,853	54 783,691	73 636,406	69,19 %	—
2025-06-25	143 294,000	39 835,732	59 316,074	0,770	0,853	51 734,717	69 538,188	69,19 %	—
2025-06-26	145 755,000	40 519,890	60 334,796	0,770	0,853	52 623,234	70 732,469	69,19 %	—
2025-06-27	141 693,000	39 390,654	58 653,345	0,770	0,853	51 156,694	68 761,249	69,19 %	—
2025-06-28	138 405,000	38 476,590	57 292,288	0,770	0,853	49 969,597	67 165,637	69,19 %	—
2025-06-30	98 117,000	27 276,526	40 615,206	0,770	0,853	35 424,060	47 614,544	69,19 %	—
Total	3 063 732,000	851 717,496	1 268 221,650	—	—	1 106 126,621	1 486 778,022	69,19 %	

Non-production days (7): 2025-06-01, 2025-06-02, 2025-06-08, 2025-06-15, 2025-06-22, 2025-06-23, 2025-06-29. These dates have no row in daily_production and are reported as N/A by omission. Weekends and public holidays in Colombia during June 2025 account for all of them.

Methodology & References

Volumetric conversion applies the canonical relation `litres = kg / density` at the calibrated product density measured at 15 °C for the reporting month. Densities are resolved per production month from the platform `product_densities` table (one row per (product, effective_from) pair), so historical conversions remain reproducible if downstream batches recalibrate the product specification. The densities applied to June 2025 are:

- **EU-grade (DEV-P100-EU)** — $\rho_{\text{EU}} = 0,770 \text{ kg/L}$
- **PLUS-grade (DEV-P100-PLUS)** — $\rho_{\text{PLUS}} = 0,853 \text{ kg/L}$

Both density values are ISCC-recognised product-specification figures for the OisteBio DEV-P100 — refined pyrolysis oil family. They are seeded and maintained in the platform `product_densities` table — initially loaded by Alembic migration `0005_seed_product_densities.py` and extended with per-month effective rows by `0014_monthly_densities.py`. Once a reporting period closes, the row applicable to that month becomes immutable for audit purposes; new effective rows are appended forward, never back-dated.

Per-row litre values shown in the daily table are not stored independently — they are derived deterministically inside PostgreSQL as **GENERATED ALWAYS** columns on the `daily_production` table:

```
litres_eu    NUMERIC GENERATED ALWAYS AS (eu_prod_kg / 0.77) STORED,
litres_plus NUMERIC GENERATED ALWAYS AS (plus_prod_kg / 0.853) STORED
```

These columns are introduced by Alembic migration `0007_persist_production_litres.py`. Once persisted, historical litre values become immutable for audit purposes even if product densities are revised in the future. Write access to `litres_eu` and `litres_plus` is prohibited at the schema level: any attempt to update them raises PostgreSQL error code 42601.

The **input kg** column reports the daily value of `daily_production.kg_to_production` — the mass of pre-conditioned ELT feedstock routed to the pyrolysis reactor that calendar day. The **yield %** column reports the fraction of input mass that exits the refining line as liquid product (EU + PLUS), with the remainder distributed across carbon black, steel scrap, water condensate, syngas, and recorded losses (totals reported in the Executive Summary above and reconciled in Annex A).

This Production Conversion Log accompanies Annex A (mass balance) and Annex D (closing-stock carry-over) in the RTFO-310825 submission bundle and is cryptographically anchored to the cover letter via the SHA-256 digest reported in the running footer (short prefix `000000000000`). The authoritative digest is reproduced in `MANIFEST.sha256` at the bundle root.

— End of Production Conversion Log —